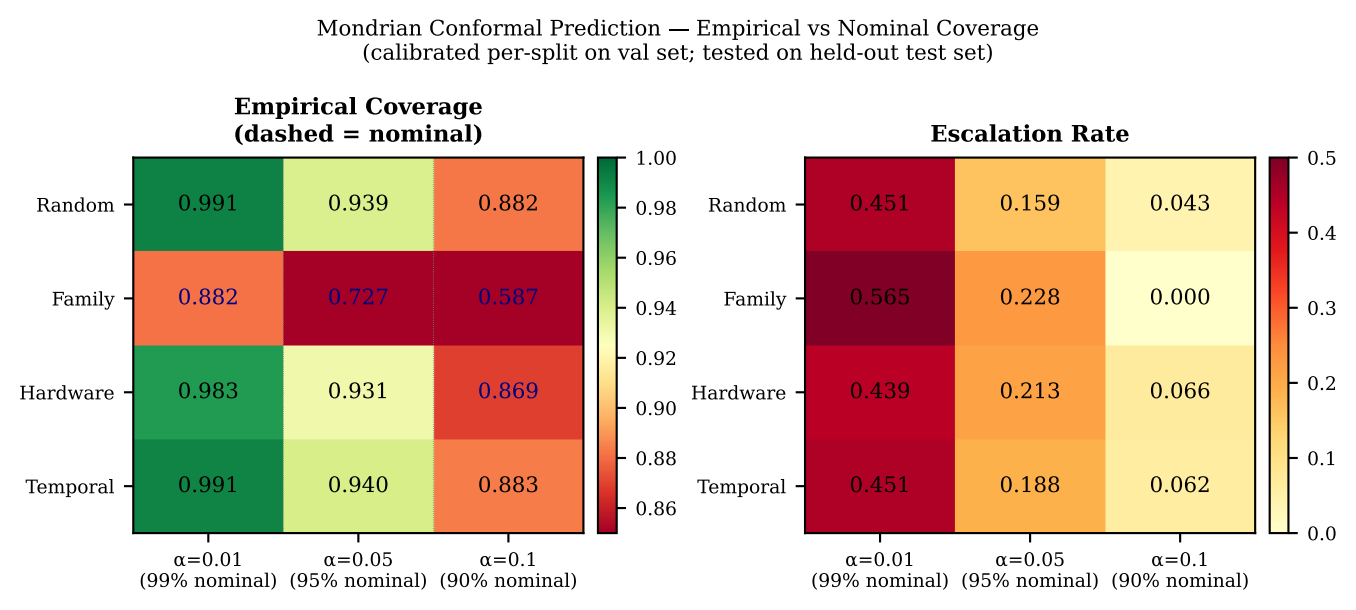
Mondrian Conformal Prediction — Empirical vs Nominal Coverage
(calibrated per-split on val set; tested on held-out test set)
Empirical Coverage
(dashed = nominal)
Random
Family
Hardware
Temporal
0.991
0.939
0.882
0.882
0.727
0.587
0.983
0.931
0.869
0.991
0.940
0.883
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)
Escalation Rate
Random
Family
Hardware
Temporal
0.451
0.159
0.043
0.565
0.228
0.000
0.439
0.213
0.066
0.451
0.188
0.062
0.5
0.4
0.3
0.2
0.1
0.0
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)